AMC 10 Crash Course · Test T1 · Algebra

10 problems · 30 minutes · no calculator · mastery bar 7 of 10

Name: ______________________ Date: ______________________

1. A price was increased by 20% twice in succession, after which it was \$144. What was the original price (in dollars)?

(A) 96 **(B)** 100 **(C)** 104 **(D)** 110 **(E)** 120

2. Numbers a and b are in the ratio $3 : 5$, and $a + b = 64$. What is $b - a$?

(A) 8 **(B)** 12 **(C)** 16 **(D)** 24 **(E)** 40

3. A cyclist rides half of a route at 20 mph and the other half at 30 mph. What is the average speed (mph)?

(A) 24 **(B)** 25 **(C)** 26 **(D)** 27 **(E)** 28

4. Numbers x and y satisfy $x + y = 10$ and $x^2 - y^2 = 40$. What is x?

(A) 4 **(B)** 5 **(C)** 6 **(D)** 7 **(E)** 8

5. One pipe fills a tank in 2 hours, another in 3. How many hours do they need together?

(A) 1 **(B)** 1.2 **(C)** 1.5 **(D)** 2.5 **(E)** 5

6. Let r and s be the roots of $x^2 - 9x + 14 = 0$. What is $r^2 + s^2$?

(A) 25 **(B)** 39 **(C)** 45 **(D)** 49 **(E)** 53

7. What is the least value of $x^2 - 8x + 3$?

(A) -16 **(B)** -13 **(C)** -8 **(D)** 3 **(E)** 13

8. A real number x satisfies $x + 1/x = 6$. What is $x^2 + 1/x^2$?

(A) 30 **(B)** 32 **(C)** 34 **(D)** 36 **(E)** 38

9. What is the sum $\frac{1}{1\cdot2} + \frac{1}{2\cdot3} + \ldots + \frac{1}{19\cdot20}$?

(A) $\frac{9}{10}$ **(B)** $\frac{18}{19}$ **(C)** $\frac{19}{20}$ **(D)** 1 **(E)** $\frac{20}{19}$

10. What is the sum of the solutions of $|2x - 5| = 9$?

(A) -14 **(B)** -5 **(C)** 2 **(D)** 5 **(E)** 14

Answer Key and Review

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1	2	3	4	5	6	7	8	9	10
B	**C**	**A**	**D**	**B**	**E**	**B**	**C**	**C**	**D**

1. $144 = x \cdot 1.2 \cdot 1.2$, so $x = 144/1.44 = 100$.

2. One part $k = 64/8 = 8$: the numbers are 24 and 40, difference 16.

3. $2 \cdot 20 \cdot 30/(20+30) = 24$. Not 25: the slower half takes more time.

4. $x^2 - y^2 = (x+y)(x-y) = 40$, so $x - y = 4$ and $x = 7$.

5. $1/2 + 1/3 = 5/6$ of the tank per hour: $6/5 = 1.2$ hours.

6. Vieta: $81 - 2 \cdot 14 = 53$.

7. Vertex at $x = 4$: $16 - 32 + 3 = -13$.

8. $36 - 2 = 34$.

9. Telescoping: $1 - 1/20 = 19/20$.

10. The solutions are 7 and -2; their sum is 5.